> $eq1 := diff(x(t), t\$4) - x(t) = 0$

$$eq1 := \frac{d^4}{dt^4} x(t) - x(t) = 0 \quad (1)$$

> $dsolve(eq1)$

$$x(t) = c_1 e^{t} + c_2 e^{-t} + c_3 \sin(t) + c_4 \cos(t) \quad (2)$$

> $eq2 := diff(x(t), t) + t \cdot x(t) = 0$

$$eq2 := \frac{d}{dt} x(t) + t\, x(t) = 0 \quad (3)$$

> $dsolve(eq2)$

$$x(t) = c_1 e^{-\frac{t2}{2}} \quad (4)$$

> $eq3 := diff(x(t), t\$2) + x(t) = 0$

$$eq3 := \frac{d^2}{dt^2} x(t) + x(t) = 0 \quad (5)$$

> $dsolve(eq3)$

$$x(t) = c_1 \sin(t) + c_2 \cos(t) \quad (6)$$

> $eq4 := 4 \cdot diff(x(t), t\$2) + 8 \cdot diff(x(t), t) + 5 \cdot x(t) = 0$

$$eq4 := 4 \frac{d^2}{dt^2} x(t) + 8 \frac{d}{dt} x(t) + 5\, x(t) = 0 \quad (7)$$

> $dsolve(eq4)$

$$x(t) = c_1 e^{-t} \sin\left(\frac{t}{2}\right) + c_2 e^{-t} \cos\left(\frac{t}{2}\right) \quad (8)$$

> $eq5 := diff(x(t), t\$2) - 3 \cdot diff(x(t), t) + 2 \cdot x(t) = 0$

$$eq5 := \frac{d^2}{dt^2} x(t) - 3 \frac{d}{dt} x(t) + 2\, x(t) = 0 \quad (9)$$

> $dsolve(eq5)$

$$x(t) = c_1 e^{t} + c_2 e^{2t} \quad (10)$$

> $eq6 := diff(x(t), t\$2) + x(t) = 0$

$$eq6 := \frac{d^2}{dt^2} x(t) + x(t) = 0 \quad (11)$$

> $ic6 := x\left(\frac{Pi}{2}\right) = 1, D(x)\left(\frac{Pi}{2}\right) = -2$

$$ic6 := x\left(\frac{\pi}{2}\right) = 1, D(x)\left(\frac{\pi}{2}\right) = -2 \quad (12)$$

> $sol6 := dsolve(\{eq6, ic6\}, x(t))$

$$sol6 := x(t) = \sin(t) + 2\cos(t) \quad (13)$$

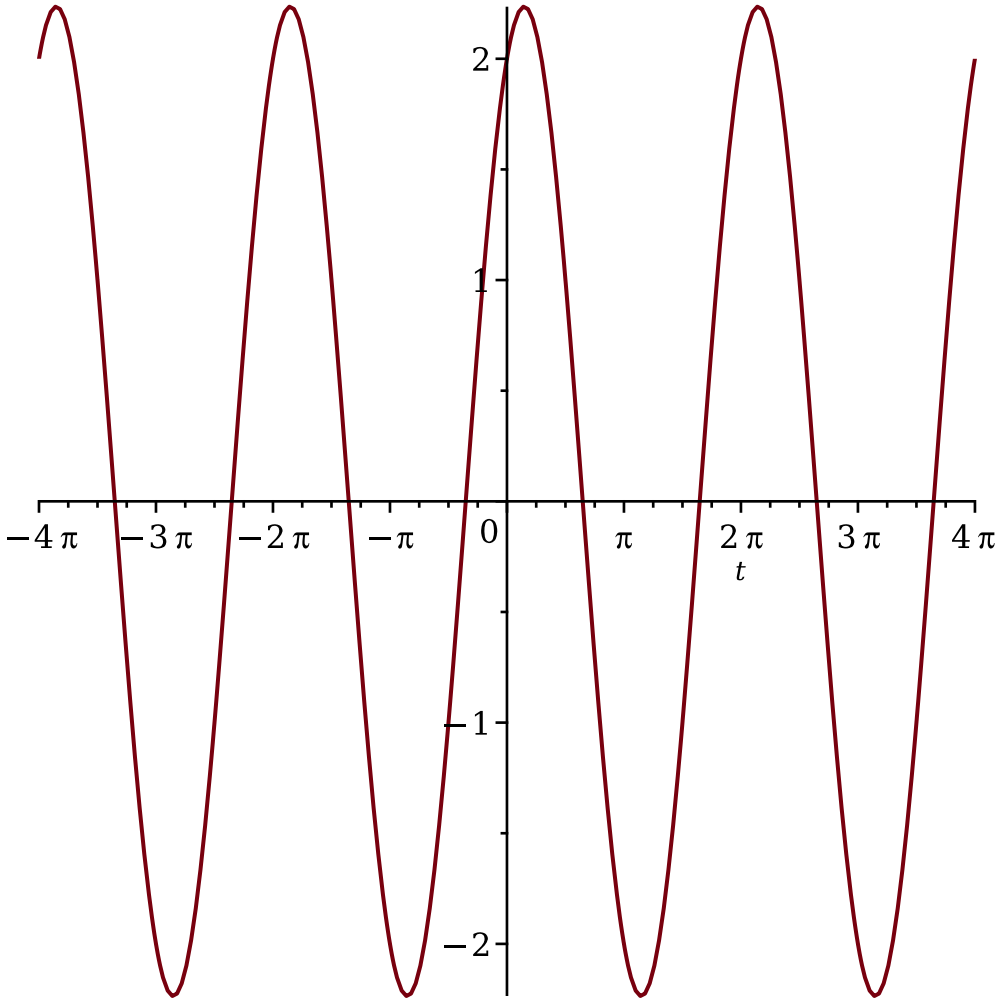
> $expand\left(sqrt(5) \cdot \cos\left(t - \arctan\left(\frac{1}{2}\right)\right)\right)$

$$\sin(t) + 2\cos(t) \quad (14)$$

```
> with(plots)
```

[animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot] **(15)**

```
> plot(rhs(sol6), t = -4·Pi..4·Pi)
```




```
> eq7 := 4·diff(x(t), t$2) + diff(x(t), t)·8 + 5·x(t) = 0
```

$$eq7 := 4\,\frac{d^2}{dt^2}x(t) + 8\,\frac{d}{dt}x(t) + 5\,x(t) = 0 \quad \textbf{(16)}$$

```
> ic7 := x(0) = 0, D(x)(0) = 0.5
```

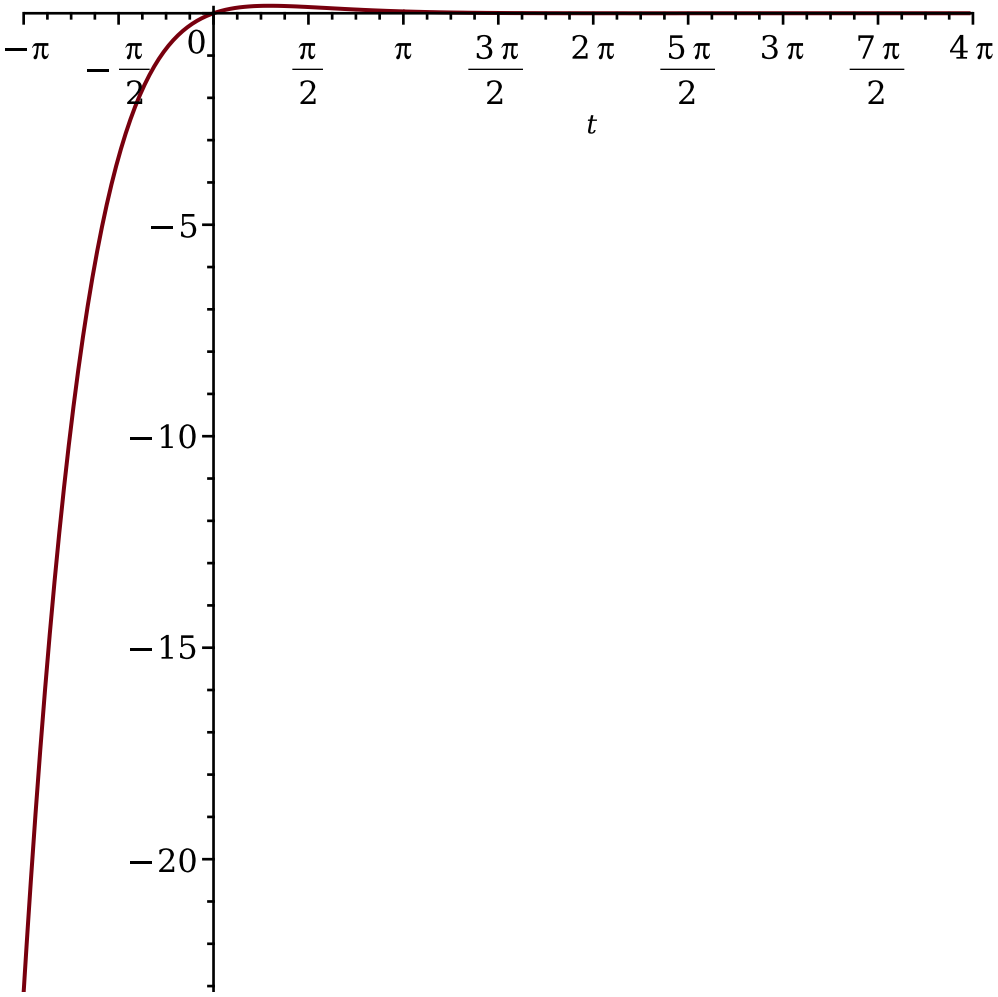
(17)

$$ic7 := x(0) = 0,\ D(x)(0) = 0.5 \tag{17}$$

> $sol7 := dsolve(\{eq7, ic7\}, x(t))$

$$sol7 := x(t) = e^{-t} \sin\left(\frac{t}{2}\right) \tag{18}$$

> $plot(\mathrm{rhs}(sol7), t = -\mathrm{Pi}..4\cdot\mathrm{Pi})$



> $eq8 := diff(x(t), t\$2) - 3\cdot diff(x(t), t) + 2\cdot x(t) = 0$

$$eq8 := \frac{d^2}{dt^2} x(t) - 3\frac{d}{dt} x(t) + 2x(t) = 0 \tag{19}$$

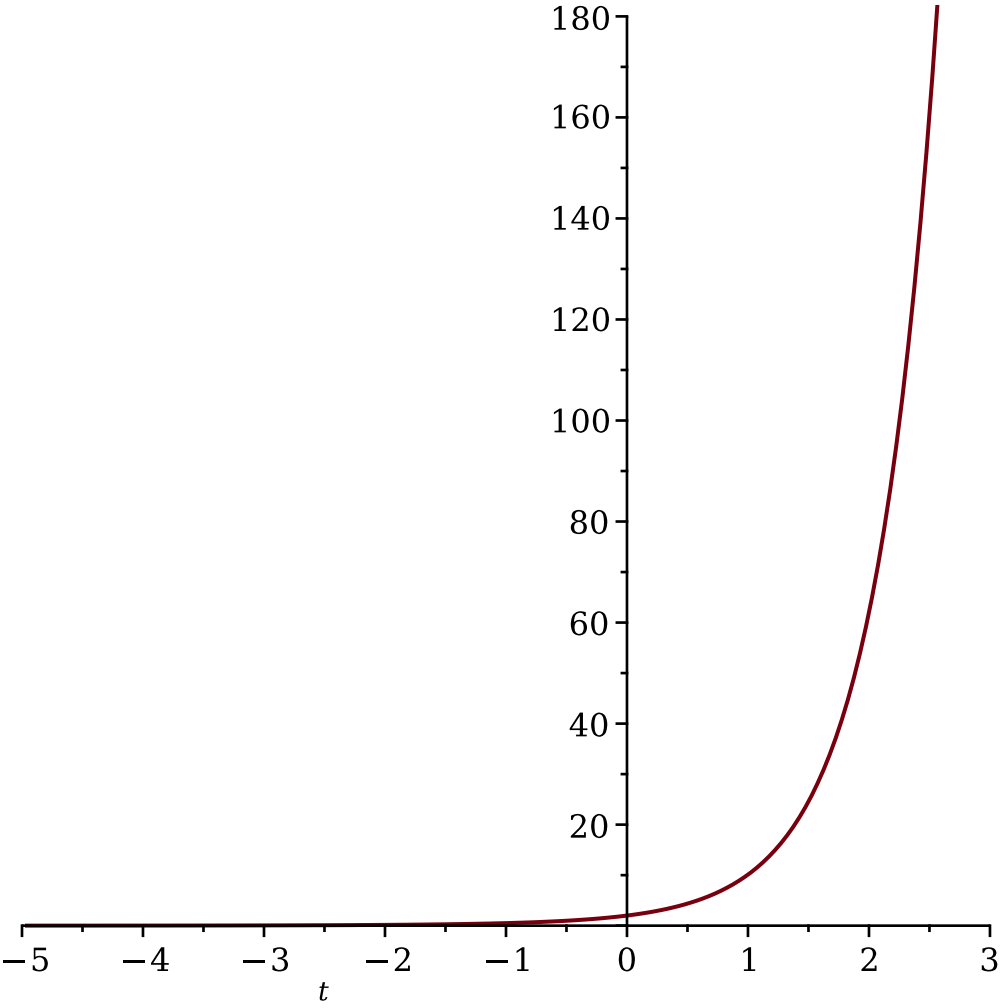
> $ic8 := x(0) = 2,\ D(x)(0) = 3$

$$ic8 := x(0) = 2,\ D(x)(0) = 3 \tag{20}$$

> $sol8 := dsolve(\{eq8, ic8\}, x(t))$

$$sol8 := x(t) = e^{t} + e^{2t} \tag{21}$$

> $plot(\mathrm{rhs}(sol8), t = -5..5)$



> $eq9 := diff(x(t), t\$2) + 5 \cdot x(t) = 0$

$$eq9 := \frac{d^2}{dt^2} x(t) + 5\, x(t) = 0 \quad (22)$$

> $dsolve(eq9, x(t))$

$$x(t) = c_1 \sin\left(\sqrt{5}\, t\right) + c_2 \cos\left(\sqrt{5}\, t\right) \quad (23)$$

> $eq10 := diff(x(t), t\$2) + t \cdot x(t) = 0$

$$eq10 := \frac{d^2}{dt^2} x(t) + t\, x(t) = 0 \quad (24)$$

> $dsolve(eq10, x(t))$

$$x(t) = c_1\, \mathrm{AiryAi}(-t) + c_2\, \mathrm{AiryBi}(-t) \quad (25)$$

> $eq11 := diff(x(t), t\$2) + t^5 \cdot x(t) = 0$

$$eq11 := \frac{d^2}{dt^2} x(t) + t^5\, x(t) = 0 \quad (26)$$

> $dsolve(eq11, x(t))$

$$x(t) = c_1 \sqrt{t}\, \mathrm{BesselJ}\left(\frac{1}{7}, \frac{2\, t^{7/2}}{7}\right) + c_2 \sqrt{t}\, \mathrm{BesselY}\left(\frac{1}{7}, \frac{2\, t^{7/2}}{7}\right) \quad (27)$$

> $infolevel[dsolve] := 3$

$$infolevel_{dsolve} := 3 \tag{28}$$

```
> dsolve(eq9, x(t))
```

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

$$x(t) = c_1 \sin\left(\sqrt{5}\, t\right) + c_2 \cos\left(\sqrt{5}\, t\right) \tag{29}$$

```
> dsolve(eq10, x(t))
```

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing 'y'
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
   -> Bessel
   <- Bessel successful
<- special function solution successful
```

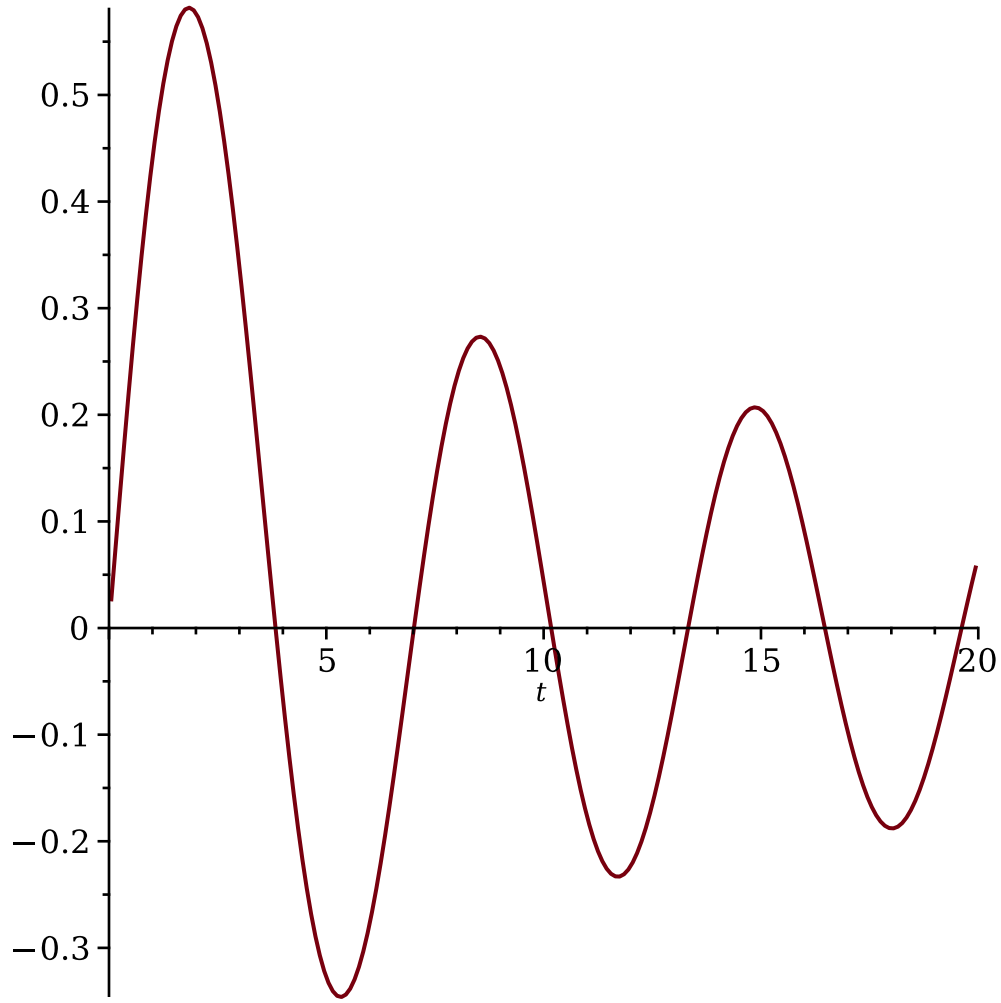
$$x(t) = c_1 \,\mathrm{AiryAi}(-t) + c_2 \,\mathrm{AiryBi}(-t) \tag{30}$$

```
> dsolve(eq11, x(t))
```

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing 'y'
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
   -> Bessel
   <- Bessel successful
<- special function solution successful
```

$$x(t) = c_1 \sqrt{t}\, \mathrm{BesselJ}\left(\frac{1}{7}, \frac{2\, t^{7/2}}{7}\right) + c_2 \sqrt{t}\, \mathrm{BesselY}\left(\frac{1}{7}, \frac{2\, t^{7/2}}{7}\right) \tag{31}$$

```
> plot(BesselJ(1, t), t = 0 .. 20)
```



> $infolevel[dsolve] := 1$

$$infolevel_{dsolve} := 1 \quad (32)$$

> $eq94 := diff(x(t), t\$2) + a(t)\cdot x(t) = 0$

$$eq94 := \frac{d^2}{dt^2} x(t) + a(t)\, x(t) = 0 \quad (33)$$

> $ic94 := x(0) = 0, D(x)(0) = 0$

$$ic94 := x(0) = 0, D(x)(0) = 0 \quad (34)$$

> $dsolve(\{eq94, ic94\}, x(t))$

> $eq95 := diff(x(t), t\$2) + 5\cdot x(t) = 0$

$$eq95 := \frac{d^2}{dt^2} x(t) + 5\, x(t) = 0 \quad (35)$$

> $dsolve(\{eq95, ic94\}, x(t))$

$$x(t) = 0 \quad (36)$$

> $eq96 := diff(x(t), t\$2) + t\cdot x(t) = 0$

$$eq96 := \frac{d^2}{dt^2} x(t) + t\, x(t) = 0 \quad (37)$$

> $dsolve(\{eq96, ic94\}, x(t))$

<- No Liouvillian solutions exists

$$x(t) = 0 \tag{38}$$

> $eq97 := diff(x(t), t\$2) + t^5 \cdot x(t) = 0$

$$eq97 := \frac{d^2}{dt^2} x(t) + t^5 x(t) = 0 \tag{39}$$

> $dsolve(\{eq97, ic94\}, x(t))$

<- No Liouvillian solutions exists

$$x(t) = c_1 \sqrt{t}\ \mathrm{BesselJ}\left(\frac{1}{7}, \frac{2 t^{7/2}}{7}\right) \tag{40}$$

> $eq10 := diff(x(t), t\$2) + x(t) = 0$

$$eq10 := \frac{d^2}{dt^2} x(t) + x(t) = 0 \tag{41}$$

> $bc10 := x(0) = 0, x(\mathrm{Pi}) = 0$

$$bc10 := x(0) = 0, x(\pi) = 0 \tag{42}$$

> $dsolve(\{eq10, bc10\}, x(t))$

$$x(t) = c_1 \sin(t) \tag{43}$$

> $eq11 := diff(x(t), t\$2) + x(t) = 0$

$$eq11 := \frac{d^2}{dt^2} x(t) + x(t) = 0 \tag{44}$$

> $bc11 := x(0) = 0, x(1) = 0$

$$bc11 := x(0) = 0, x(1) = 0 \tag{45}$$

> $dsolve(\{eq11, bc11\}, x(t))$

$$x(t) = 0 \tag{46}$$

> $eq12 := diff(x(t), t\$2) + x(t) = 1$

$$eq12 := \frac{d^2}{dt^2} x(t) + x(t) = 1 \tag{47}$$

> $bc12 := x(0) = 0, x(\mathrm{Pi}) = 0$

$$bc12 := x(0) = 0, x(\pi) = 0 \tag{48}$$

> $dsolve(\{eq12, bc12\}, x(t))$

> $dsolve(diff(x(t), t) + x(t) = 15, x(t))$

$$x(t) = 15 + e^{-t} c_1 \tag{49}$$

> $sol13 := dsolve(diff(x(t), t) + x(t) = 2 \cdot \exp(t) - 7 \cdot \exp(-3 \cdot t))$

$$sol13 := x(t) = \frac{7 e^{-3t}}{2} + e^{t} + e^{-t} c_1 \tag{50}$$

> $subs(_C1 = 0, sol13)$

$$x(t) = \frac{7 e^{-3t}}{2} + e^{t} \tag{51}$$

> $sol14 := dsolve(diff(x(t), t) + x(t) = 2 \cdot \exp(t) - 7 \cdot \exp(-3 \cdot t))$

$$sol14 := x(t) = \frac{7 e^{-3t}}{2} + e^{t} + e^{-t} c_1 \tag{52}$$

```
> subs(_C1 = 0, sol14)
```

$$x(t) = \frac{7\,e^{-3\,t}}{2} + e^{t} \tag{53}$$

```
> sol15 := dsolve(diff(x(t), t) + x(t) = -t^2 + 3·t - 7)
```

$$sol15 := x(t) = -t^2 + 5\,t - 12 + e^{-t}\,c_1 \tag{54}$$

```
> subs(_C1 = 0, sol15)
```

$$x(t) = -t^2 + 5\,t - 12 \tag{55}$$

```
> sol16 := dsolve(diff(x(t), t) + x(t) = sin(t) + 3·cos(t))
```

$$sol16 := x(t) = \cos(t) + 2\sin(t) + e^{-t}\,c_1 \tag{56}$$

```
> subs(_C1 = 0, sol16)
```

$$x(t) = \cos(t) + 2\sin(t) \tag{57}$$

```
> sol17 := dsolve(diff(x(t), t) + x(t) = sin(t))
```

$$sol17 := x(t) = -\frac{\cos(t)}{2} + \frac{\sin(t)}{2} + e^{-t}\,c_1 \tag{58}$$

```
> subs(_C1 = 0, sol17)
```

$$x(t) = -\frac{\cos(t)}{2} + \frac{\sin(t)}{2} \tag{59}$$

```
> sol18 := dsolve(diff(x(t), t) + x(t) = 3*cos(t))
```

$$sol18 := x(t) = \frac{3\cos(t)}{2} + \frac{3\sin(t)}{2} + e^{-t}\,c_1 \tag{60}$$

```
> subs(_C1 = 0, sol18)
```

$$x(t) = \frac{3\cos(t)}{2} + \frac{3\sin(t)}{2} \tag{61}$$

```
>
```